

Math 675 Spring 2026:
Homework 4 (LATE)
Due Tuesday, April 6th 2026

Question 1:

Consider the matrix $A = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix}$

The eigenvalues for A are $\lambda_1 = -1$ and $\lambda_2 = -6$ with corresponding eigenvectors $\mathbf{v}_{\lambda_1} = (1, 4)$ and $\mathbf{v}_{\lambda_2} = (-1, 1)$. Use this and the method of undetermined coefficients to find the general solution to the inhomogeneous system:

$$\mathbf{x}'(t) = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix} \mathbf{x}(t) + e^{2t} \begin{pmatrix} 6 \\ -1 \end{pmatrix}$$

Question 2:

Consider the matrix $A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$

The eigenvalues for A are $\lambda_1 = -1$ and $\lambda_2 = 1$ with corresponding eigenvectors $\mathbf{v}_{\lambda_1} = (1, 3)$ and $\mathbf{v}_{\lambda_2} = (1, 1)$.

1. Calculate $\exp(tA)$

Given the definition: $\exp(tA) = \Psi(t)\Psi^{-1}(0)$, where $\Psi(t)$ is the fundamental matrix made up of the x and y solution vectors from $\mathbf{x}(t)$.

First, build the general solution:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Psi(t) = \begin{pmatrix} e^{-t} & e^t \\ 3e^{-t} & e^t \end{pmatrix}$$

$$\Psi^{-1}(0) = (\Psi(0))^{-1} = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\exp(tA) = \Psi(t)\Psi^{-1}(0) = \begin{pmatrix} e^{-t} & e^t \\ 3e^{-t} & e^t \end{pmatrix} \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\exp(tA) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t & e^{-t} - e^t \\ -3e^{-t} + 3e^t & 3e^{-t} - e^t \end{pmatrix}$$

Now, optionally, we could convert this into a form involving hyperbolic trig:

$$\exp(tA) = \begin{pmatrix} \sinh(t) & -\sinh(t) \\ 3\sinh(t) & \cosh(t) - 2\sinh(t) \end{pmatrix}$$

2. Solve the IVP

Use the result from part (1) to solve:

$$\mathbf{x}'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Defintion: Given a system $\mathbf{x}'(t) = A\mathbf{x}(t)$ the solution is $\mathbf{x}(t) = e^{tA}\mathbf{x}(0)$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t & e^{-t} - e^t \\ -3e^{-t} + 3e^t & 3e^{-t} - e^t \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t + 2e^{-t} - 2e^t \\ -3e^{-t} + 3e^t + 6e^{-t} - 2e^t \end{pmatrix}$$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} e^{-t} + e^t \\ 3e^{-t} + e^t \end{pmatrix}$$

$$\mathbf{x}(t) = \frac{1}{2}e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \frac{1}{2}e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Ta-Da! This is the same IVP Solution as if we solved this the regular way!

Question 3:

The following questions refer to the proof of existence and uniqueness covered in chapter 2.3 of the guided notes.

a) Why do we choose a small time interval?

In the proof we chose $\alpha < \min \left\{ \frac{\rho}{M}, \frac{1}{B} \right\}$

(a) Explain in words why we need the condition $\alpha < \frac{\rho}{M}$

We need this in order to bound the first iteration by ρ similarly as ε

$$\|x_1(t) - x_0\| = \|t\| \cdot \|\mathbf{F}(x_0)\| \leq \alpha M \leq \frac{\rho}{M} \cdot M \leq \rho$$

This makes sure that $\forall k, x_k$ remains inside $\mathcal{O}_\rho$ centered about x_0

- (b) Explain in words why we need the condition $\alpha < \frac{1}{B}$
 We need this in order for the normed difference of sequential iterations to be less than epsilon for convergence. So the $(\alpha B)^k$ term below can get small as we let k be big, which means the iterations get closer together

$$\|\mathbf{x}_{k+1}(t) - \mathbf{x}_k(t)\| \leq (\alpha B)^k \cdot L$$

Later on, we use $\gamma = \alpha B < 1$ so that we can build a convergent geometric series to compare with the normed telescoping series of iterations

- (c) What could go wrong if α was too large?
 This could cause x_k to leave $\mathcal{O}_\rho$, which may not guarantee convergence for the ODE solution, and the geometric series $\sum \gamma$ may not converge

b) Staying inside the closed ball

Suppose we have $x_{k+1}(t) = x_0 + \int_0^t F(x_k(s)) ds$

- (a) Why is it important that all iterates stay inside $\mathcal{O}_\rho$
 If we stay $\mathcal{O}_\rho$, we then have keep the Lipschitz condition true, we can bound the slope of $\mathbf{F}$ on the domain
- (b) What assumptions about F would fail if an iterate left $\mathcal{O}_\rho$?
 If $x' \notin \mathcal{O}_\rho$ then $\|\mathbf{F}(x')\|$ would not be bounded by M
- (c) Explain how the bound $\|F(w)\| \leq M$ is used in this step
 Like before, we can use this to bound the first normed difference and the following terms recursively

$$\|x_1(t) - x_0\| = \|t\| \cdot \|\mathbf{F}(x_0)\| \leq \alpha M \leq \frac{\rho}{M} \cdot M \leq \rho$$

c) Role of Lipschitz Condition

The proof uses $\|F(x) - F(y)\| \leq B\|x - y\|$

- (a) Where in the proof is this condition used?

$$\begin{aligned} \|\mathbf{x}_2(t) - \mathbf{x}_1(t)\| &\leq \int_0^t \|\mathbf{F}(\mathbf{x}_1(s)) - \mathbf{F}(\mathbf{x}_0(s))\| ds \\ &\leq \int_0^t B\|\mathbf{x}_1(s) - \mathbf{x}_0\| ds \leq B \cdot t \cdot L \leq \alpha \cdot B \cdot L \end{aligned}$$

- (b) Why is continuity of $\mathbf{F}$ alone not enough for the argument?
 Would not guarantee that the slope of $\mathbf{F}$ won't be too large or blow up. The best we would be able to do is bound $\|\mathbf{F}(x_{k+1}) - \mathbf{F}(x_k)\|$ by $2M$ by triangle inequality, yet we also have no guarantee that $\sum (\alpha M)^k$ would not blow up to infinity.

(c) In your own words, what does the Lipschitz condition guarantee?

This condition allows us to always bound $\|x_{k+1} - x_k\|$ by the $(\alpha B)^k L$ term. Which, as $k \rightarrow \infty$, then $\|x_{k+1} - x_k\| \rightarrow 0$

d) **Why does a geometric series appear?**

We obtained an estimate $\|x_{k+1}(t) - x_k(t)\| \leq (\alpha B)^k L$

(a) Why is it important that $\alpha B < 1$?

So that the geometric series $\sum (\alpha B)^k L \leq \sum (\gamma)^k L$ converges

(b) What would happen if $\alpha B \geq 1$?

This would result in $\sum (\alpha B)^k$ not converging and thus $\sum \|x_{k+1} - x_k\| \not\leq \varepsilon$, meaning the need lemma for convergence could not be applied

(c) How does this estimate lead to convergence of $\{x_k\}$

$$\begin{aligned} \sum_{k=N}^{\infty} \|x_{k+1} - x_k\| &= \|x_{N+1} - x_N\| + \|x_{N+2} - x_{N+1}\| + \dots \\ &\leq (\alpha B)^N L + (\alpha B)^{N+1} L + \dots = \sum_{k=N}^{\infty} (\alpha B)^k L < \varepsilon \end{aligned}$$

since $(\alpha B) < 1$, RHS converges by Cauchy Conditions, so LHS does too. and since LHS sum converges to less than epsilon, then there exists a function that $\{x_k\}$ converges to. So we now have $\{x_k(t)\} \rightarrow x(t)$

e) **Conceptual comparison**

Compare:

- Solving an algebraic equation $x = G(x)$ via fixed point iteration
- Solving the integral equation $x(t) = x_0 + \int_0^t F(x(s))ds$

(a) What plays the role of the function being iterated?

With Fixed point iteration, we want to obtain a single value for x^* that gets iterated from x_k . But in Picard iteration, we want to obtain some sized vector $\vec{x}^*$ from iterating that same sized vector $\vec{x}_k$

(b) What plays the role of the fixed point?

Like above, we want x^* or $\vec{x}^*$ as the fixed point as $k \rightarrow \infty$

(c) Why is this analogy useful?

This allows us to connect multiple fields in math that we have learned. Like how a fixed point problem could be turned into a root finding problem. Now an system of ODE became a fixed point problem. Yet also gives us a better conceptual of whats actually going on behind the scenes.